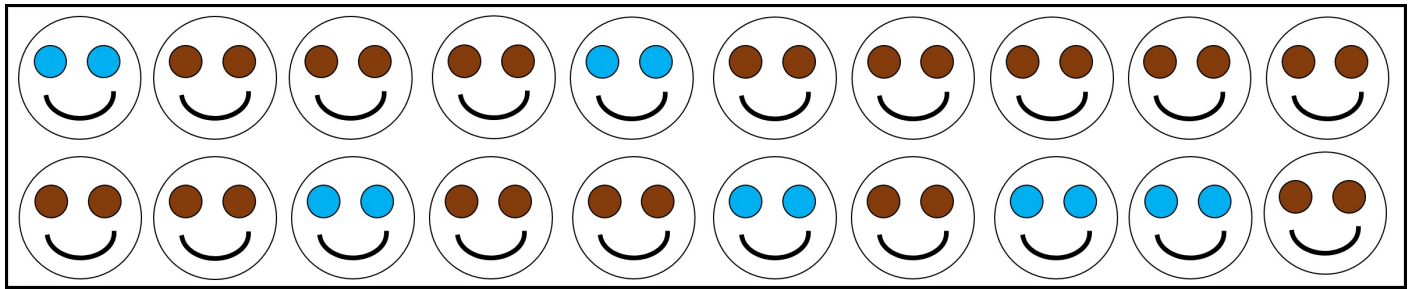


PROBLEM #1.

You have sampled a population for their eye color and only two phenotypes exist, BROWN (which is dominant) and BLUE (which is recessive). If the brown allele (“A”) is dominant over the recessive allele (“a”) and these are the ONLY two alleles in the population, answer the following questions. The data set for the population is shown below. Your answers should be rounded to one decimal place and expressed as a percentage (i.e. 0.4768 would be 47.7%). [NOTE: the answers below are expanded and worked out in with detailed explanations].



A. **The frequency of the recessive phenotype.**

The recessive phenotype (physical character) would be the BLUE eyed condition. Based on the data set provided above, 6 out of 20 people have blue eyes, or **30.0%**

B. **The frequency of the dominant phenotype.**

The dominant phenotype (physical character) would be the BROWN eyed condition. Based on the data set provided above, 14 out of 20 people have blue eyes, or **70.0%**. Remember, the sum of BOTH phenotypes (blue and brown) MUST equal 100.0%, which they do in this case.

C. **The frequency of the “a” allele.**

The frequency of the “a” allele (which is the recessive allele) would correspond to the variable “q” in the Hardy Weinberg equation) which is $p^2 + 2pq + q^2 = 1.0$ and we also know that $p + q = 1.0$ for the equation. The variables in the HW equation are as follows:

p^2	=	% Individuals who are HOMOZYGOUS DOMINANT (“AA”, BROWN eyes)
p	=	% of the DOMINANT ALLELE (“A”, the BROWN allele)
$2pq$	=	% Individuals who are HETEROZYGOUS (“Aa”, BROWN eyes)
q^2	=	% Individuals who are HOMOZYGOUS RECESSIVE (“aa”, BLUE eyes)
q	=	% of the RECESSIVE ALLELE (“a”, the BLUE allele)
1.0	=	100% of the population or the sum of all individuals or the total of all alleles (brown + blue)

Based on the data set we have, 6/20 or 30.0% have BLUE eyes. So, q^2 would equal 0.300. To find q, we take the square root of q^2 , which would be 0.5477 or 0.548 or 54.8%. That means 54.8% of the alleles present in the sample are “a”. This also means that **q = 0.5477 or 0.548 or 54.8%**

D. **The frequency of the “A” allele.**

If 54.8% of the alleles are “a” (from above), then $1 - 0.548$ would be the “A” alleles. That means 45.2% of the alleles present in the sample are “A”. This also means that **$p = 0.4523$ or 0.452 or 45.2%** . Remember, $p + q$ must equal 1.0 (or 100%). For this question, p (45.2%) + q (54.8%) does equal 100%.

E. **The frequency of the people who have BROWN eyes that are homozygous for their genotype.**

We know from the data set that 70.0% of the people have BROWN eyes, but some are homozygous dominant (“AA”, which is p^2 in the HW equation) and some are heterozygous (“Aa”, which is $2pq$ in the HW equation).

We determined that $p = 0.4523$ and $q = 0.5477$, so p^2 would be $(0.4523) \times (0.4523) = 0.2045$ or 0.205 or **20.5% would be AA (homozygous dominant) brown eyes.**

F. **The frequency of the people who have BROWN eyes that are heterozygous for their genotype.**

We determined that $p = 0.4523$ and $q = 0.5477$, so $2pq$ would be calculated as $2 \times (0.4523) \times (0.5477) = 0.4954$ or 0.495 or **49.5% would be Aa (heterozygous) brown eyes.**

As a quick check, you can plug the values you calculated into the HW equation and make sure they sum to 1.0 or 100.0%. If we do that, then $0.205 + 0.495 + 0.300$ does equal 1.000 or 100.0%

PROBLEM #2.

Sickle-cell anemia is an interesting genetic disease. Normal homozygous individuals (SS) have normal blood cells that are easily infected with the malarial parasite. Thus, many of these individuals become very ill from the parasite and many die. Individuals homozygous for the sickle-cell trait (ss) have red blood cells that readily collapse when deoxygenated. Although malaria cannot grow in these red blood cells, individuals often die because of the genetic defect. However, individuals with the heterozygous condition (Ss) have some sickling of red blood cells, but generally not enough to cause mortality. In addition, malaria cannot survive well within these “partially defective” red blood cells. Thus, heterozygotes tend to survive better than either of the homozygous conditions.

If 9% of an African population is born with a severe form of sickle-cell anemia (ss), then answer these questions.

A. **What percentage of the population will be more resistant to malaria because they are heterozygous (Ss) for the sickle-cell gene?**

$9\% = 0.09 = ss = q^2$. To find q , simply take the square root of 0.09 to get 0.3. Since $p = 1 - 0.3$, then p must equal 0.7. $2pq = 2 (0.7 \times 0.3) = 0.42 =$ **42.0% of the population are heterozygotes (carriers).**

PROBLEM #3.

Within a population of butterflies, the color brown (B) is dominant over the color white (b). And, 40% of all butterflies are white. Given this simple information, which is something that is very likely to be on an exam, calculate the following:

- A. **The percentage of butterflies in the population that are heterozygous.**
- B. **The frequency of homozygous dominant individuals.**

Answers: The first thing you'll need to do is obtain p and q. So, since white is recessive (i.e. bb), and 40% of the butterflies are white, then $bb = q^2 = 0.4$. To determine q, which is the frequency of the recessive allele in the population, simply take the square root of q^2 which works out to be 0.632 (i.e. $0.632 \times 0.632 = 0.4$). So, $q = 0.632$. Since $p + q = 1$, then p must be $1 - 0.632 = 0.368$. Now then, to answer our questions. First, what is the percentage of butterflies in the population that are heterozygous? Well, that would be $2pq$ so the answer is $2 (0.368) (0.632) = 0.465$ or **46.5% are heterozygous**. Second, what is the frequency of homozygous dominant individuals? That would be p^2 or $(0.368)^2 = 0.135$ or **13.5% are homozygous dominant**.

PROBLEM #4.

A rather large population of organisms have 396 red-sided individuals and 557 tan-sided individuals. Assume that red is totally recessive. Please calculate the following:

- A. The allele frequencies of each allele. Answer: Well, before you start, note that the allelic frequencies are p and q, and be sure to note that we don't have nice round numbers and the total number of individuals counted is $396 + 557 = 953$. So, the recessive individuals are all red (q^2) and $396/953 = 0.416$. Therefore, q (the square root of q^2) is 0.645. Since $p + q = 1$, then p must equal $1 - 0.645 = 0.355$. So, the **tan allele frequency is 35.5%** and the **red allele frequency is 64.5%** in the population.
- B. The expected genotype frequencies. Answer: Well, $AA = p^2 = (0.355)^2 = 0.126$; $Aa = 2(p)(q) = 2(0.355)(0.645) = 0.458$; and finally $aa = q^2 = (0.645)^2 = 0.416$ (you already knew this from part A above). **AA (homozygous dominant TAN) is 12.6%**, **Aa (heterozygous TAN) is 45.8%**, and **aa (homozygous recessive RED) is 41.6%** for the population.
- C. The number of heterozygous individuals that you would predict to be in this population. Answer: That would be $0.458 \times 953 =$ about **436**.
- D. The expected phenotype frequencies. Answer: Well, the "A" phenotype = $0.126 + 0.458 = 0.584$ or **58.4% are TAN**. The "a" phenotype = 0.416 or **41.6% are RED** (but you already knew these numbers from the calculations you did in PART A above..
- E. Conditions happen to be really good this year for breeding and next year there are 1,245 offspring. Assuming that all of the Hardy-Weinberg conditions are met, how many of these would you expect to be red-sided and how many tan-sided? Answer: Simply put, The "A" phenotype = $0.584 \times 1,245 =$ **727 tan-sided** and the "a" phenotype = $0.416 \times 1,245 =$ **518 red-sided** (or $1,245 - 727 = 518$).

**These are the types of problems
that you should be able to do for
EXAM 1 and also for the QUIZ on
MOODLE**